LGTM — Checkpoint Report (due 18 Oct 2026)

Team **"wo men ai AI"** · SE201 AI-Driven Software Engineering, AY2026-27 T1

Repository: https://github.com/CaesiumSG/SE201-LGTM

This report covers the four checkpoint requirements in the project specification:

- architecture in place, with stubs documented;
- a working core feature;
- the AI technique working;
- the checkpoint document itself: test results, progress against targets, scope changes, and the AI use log.

The detailed experiment history, with every failure we hit and how we fixed it, is in `evaluation-log.md`. The evaluation on published CVEs in real open-source projects is in `real-world-eval.md`, and the prompt-injection red-teaming is in `prompt-injection.md`.

Summary

- **Status:** all architecture components are built, connected and evaluated on three separate benchmarks: synthetic (Shoply), real-world CVEs, and prompt-injection attacks.
- **Three benchmarks, reported separately.** Each answers a different question with its own script and metric, so we don't pool them into one score: Shoply's dev split was used for tuning while the other sets were never tuned on, and injection is measured by attack success rate, not F1.

Benchmark	Question	Cases	Script	Metric
Shoply	Does LGTM catch flaws without false alarms?	36 (dev 18 / eval 18)	`benchmark/evaluate.py`	F1
Real-world CVEs	Does it hold on shipped code?	12, held out	`benchmark/real_world/evaluate_real.py`	F1
Prompt injection	Can PR text trick it?	6 attacks	`benchmark/injection/run_injection_bench.py`	Attack success rate

- **Results on the synthetic benchmark** (Shoply, 36 cases; held-out eval split in brackets):

Mode	F1
LGTM	**0.94** (0.89)
LLM-only baseline	0.75 (0.75)
SAST-only baseline	0.62 (0.63)

- **Results on real-world CVEs** (6 published advisories in open-source Python projects, 12 cases, held out, nothing tuned on them):

Mode	F1
LGTM	0.44

Mode	F1
LLM-only baseline	**0.80**
SAST-only baseline	0.44

LGTM did not reproduce its synthetic result on real code. SQL injection triage carried over (SQLi F1 0.80 vs 0.67 for SAST alone). Authorization detection did not (F1 0.00): in 4 of 6 cases our pattern-based endpoint detector never sent the code to the agent. See §3a.

- **Results on prompt injection** (6 attack cases; attack success rate, lower is better; only the 3 SQLi attacks count, because no mode flagged the authz originals):

Mode	Attack success rate
SAST-only baseline	0% (0/3)
LLM-only baseline	0% (0/3)
LGTM, no mitigation	**100% (3/3)**
LGTM + `--strip-comments` + `--detect-prompt-injection`	0% (0/3)

Comments and docstrings in a PR talked LGTM's triage agent out of all three SQLi detections; with the mitigations, none succeeded. See §6 and `prompt-injection.md`.

- **What the AI adds (synthetic benchmark):**
  - Without the security policy, the bare LLM catches 0 of 6 authorization flaws; LGTM catches all 6.
  - Semgrep wrongly flags 75% of safe parameterized SQL; LGTM's triage brings that to 0%.
- **What doesn't work yet:**
  - On real projects the agents only run where a regex-style detector fires, and they see only the changed file, so authorization checks that live elsewhere are invisible to them.
  - The 7B model doesn't use its tools: 0 tool calls on all 54 real-world findings, and on Shoply forcing tool use cost 6.6× latency for no gain.
  - The authz agent raises ownership warnings on 22 of 24 endpoints in the Shoply SQLi cases, which case-level scoring doesn't count.
  - Verdicts can flip on irrelevant prompt changes.

1. Architecture: in place and connected

All five stages from the proposal are implemented and run end to end from one command (`lgtm review`). Figure 1 shows the components as built:

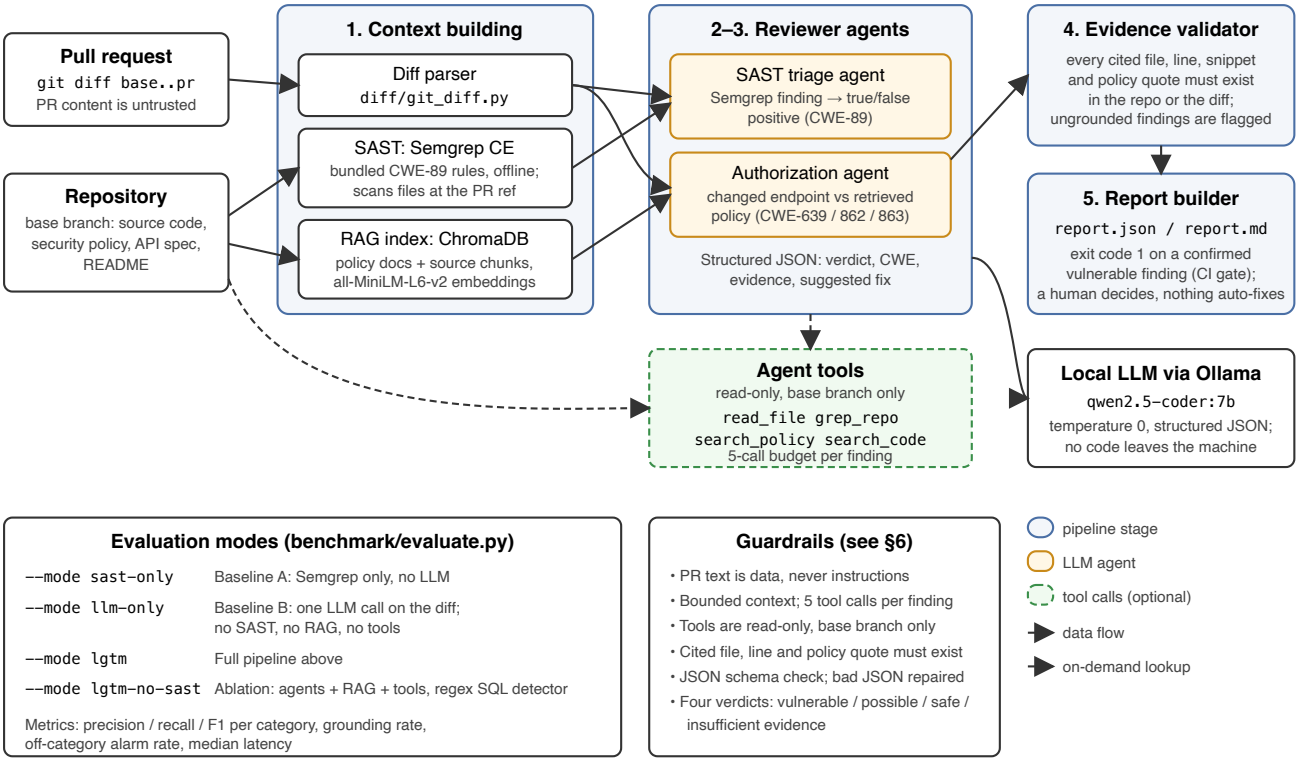


Figure 1: LGTM architecture as implemented. Solid arrows are the data flow of one review; dashed arrows are on-demand lookups the agents may make through the read-only tools. Compared with the proposal's diagram (`docs/original_architecture.drawio`), the only additions are the `search_code` tool and source code in the RAG index.

Stage	Component	Status
1. Context building	`diff/git_diff.py` (git diff), `sast/semgrep_runner.py` (Semgrep CE, bundled CWE-89 rules), `rag/` (ChromaDB index of policy docs + source code)	✅ working; the diff reader was hardened for real repos (non-UTF-8 files, quoted paths)
2. SAST triage agent	`agents/sast_triage.py` — CWE-89 true/false positive	✅ working
3. Authorization agent	`agents/authz_reviewer.py` — CWE-639/862/863 against the retrieved policy	✅ working
Agent tools	`agents/tools.py` — `read_file`, `grep_repo`, `search_policy`, `search_code`; read-only, base branch only, 5-call budget	⚠️ implemented, but rarely used by the model (see below)
4. Evidence validator	`validation/evidence.py` — cited files/lines/snippets/policy text must exist	✅ working (2 bugs fixed this week)
5. Report builder	`reporting/report_builder.py` — `report.json` + `report.md`, CWE, evidence, suggested fix	✅ working
Baselines	`--mode sast-only` (A), `--mode llm-only` (B), plus ablation `--mode lgtm-no-sast`	✅ working
Benchmark harness	`benchmark/evaluate.py` — 36 cases, dev/eval split, all proposal metrics	✅ working

Stage	Component	Status
Real-world benchmark	`benchmark/real_world/` — 12 cases from 6 published CVEs; `fetch.py` rebuilds them from the upstream fix commits, `evaluate_real.py` scores them with the same metrics	✅ working
Model server (optional)	`model/` — serves the Ollama model from the SCIS GPU cluster and tunnels it to the laptop; LGTM uses it via `--ollama-url`	✅ working; not yet used for a benchmark run

Simplified or stubbed components (honest list)

- **Tool use is effectively unused.** The tools work: they are unit-tested, and we fixed a parsing bug so that text-form tool calls are executed. But `qwen2.5-coder:7b` answers from the pre-retrieved context and almost never calls them. We tested forcing tool use as an experiment (see the evaluation log). On the real-world cases it made 0 tool calls across 54 findings. The "agentic workflow" claim should be read with this in mind.
- **SAST is SQL-injection only**, using 4 hand-written pattern rules. These are not Semgrep taint-mode rules, and not the Semgrep registry, so the tool works offline.
- **Endpoint detection is pattern-based**: single-line Flask/FastAPI route decorators, and only inside a changed hunk. The real-world evaluation showed this is the main reason LGTM misses real authorization flaws: the check is often far from the decorator, or the decorator spans several lines.
- **The authz agent sees only the endpoint file's diff.** An ownership check in another module (e.g. a CRUD layer) is not in its prompt.
- **Suggested fixes are prose, not patches.** The proposal said "propose a patch"; we output a textual fix suggestion. Patch generation is not built.
- **No GitHub integration.** LGTM is a CLI over two git refs; it does not post PR comments. A CLI is acceptable per the spec.
- **"Tokens per PR" is an estimate** (characters ÷ 4 of the findings), not the model's real token count.
- **The main benchmark is synthetic**: one small Flask app (Shoply) with 18 generated vulnerable/safe PR pairs. The real-world benchmark is real code but small (12 cases), and its vulnerable PRs are synthetic reverts of the upstream fix, not PRs a developer actually opened.
- **The OWASP cases are translated, not extracted.** OWASP Benchmark is a Java test suite; our 6 OWASP-derived pairs re-implement its SQLi patterns in Python/Flask, because LGTM targets Python.

2. Core feature demonstration

```
# one-time: index the target repo's docs and code
lgtm index --repo /path/to/repo
# review a PR branch against its base
lgtm review --repo /path/to/repo --base main --pr feature --output lgtm-output
```

The exit code is 1 when a `vulnerable` finding is reported, so the tool can gate CI. Sample outputs for every benchmark case (all findings, verdicts, evidence and validation results) are committed in `benchmark/runs/`.

3. AI technique working

- **RAG:** the security policy, API spec and README are chunked into ChromaDB. The authorization agent's prompt is seeded with the top policy chunks for the endpoint under review. Without this context the bare LLM detects **0 of 6** authorization flaws; with it, LGTM detects them (see results). This is the clearest evidence that the retrieval stage does real work.
- **SAST integration:** Semgrep findings on the changed lines seed the triage agent.
- **Structured output + guardrails:** every finding is JSON with a verdict from four options, a CWE, evidence and a fix. Malformed model JSON is repaired. Every citation is validated after the fact.
- **Tool calling:** implemented, but under-used (see §1).

3a. Real-world evaluation

Following instructor feedback, we tested LGTM on code we did not write. Full write-up: `real-world-eval.md`; cases and reproduction: `benchmark/real_world/`.

- **Cases:** 6 published advisories (GHSA/CVE) in pyload, postgraas_server, alerta (SQLi, CWE-89) and aegra, langflow, litellm (authz, CWE-639/863). Each gives a safe case (the real fix commit) and a vulnerable case (the fix reverted on top of it). Advisory ids and SHAs were checked against the GitHub API, and `fetch.py` re-verifies them on every run.
- **Protocol:** held out. All experiment flags off, and no prompt, rule or config was changed for these cases. Authz cases were run twice: with the project's own docs, and with a short policy written from its public documentation before any model run.

Mode	TP	FP	FN	TN	Precision	Recall	F1	SQLi F1	Authz F1
sast-only	2	1	4	5	0.67	0.33	0.44	0.67	0.00
llm-only	4	0	2	6	1.00	0.67	0.80	0.80	0.80
lgtm	2	1	4	5	0.67	0.33	0.44	0.80	0.00
lgtm-no-sast	2	1	4	5	0.67	0.33	0.44	0.80	0.00

What failed, and why (each cause is traced to a run file in the write-up):

1. **The agent runs only where a detector fires.** 7 of 12 LGTM cases ran no agent at all. For authz, 4 of 6 cases had no detected endpoint, because the route decorator was outside the changed lines or spanned several lines.
2. **The agent sees only the endpoint file.** In langflow the fix's ownership check is in `crud.py`, so the agent called the *fixed* code vulnerable. It made no tool call to look further.
3. **Authentication was accepted as authorization** when the retrieved text was not an access rule.
4. **The validator checks existence, not relevance.** A link sentence from `SECURITY.md`, and a policy rule about deleting *user accounts*, both passed validation and were cited as the rule for API keys.
5. **The SQLi sink pattern is narrow.** In alerta the injection is not an `execute()` call in the diff, so Semgrep and the heuristic found nothing.
6. **The LLM-only score is inflated by how the cases are built.** In two authz cases the vulnerable diff visibly *deletes* the guard, and the baseline's explanations say exactly that. A real vulnerable PR that never had the check would not show this.

What carried over: SQLi triage still removed the Semgrep false alarm (pyload), and every case with a real `execute()` sink was caught.

Learning. On Shoply, decorator and check sit together in one file, and our pre-retrieval design worked. On real projects, the context the reviewer needs is spread across files, which is the scenario where the agent should fetch code itself. It didn't: our detectors skip the agent before it can use its tools, and the 7B model doesn't call them anyway. Next steps are to let the agent run when no endpoint is detected in the diff, to put the relevant callees in its context, and to test a larger model (via the GPU model server) to see whether tool use improves.

4. Progress against the proposal's checkpoint targets

Target (from proposal §6)	Status	Notes
Runnable end-to-end CLI on an authorization PR	✅ Met	`lgtm review` from git diff to JSON finding
Semgrep integration	✅ Met	Bundled offline rules; scans PR-ref file contents
Policy retrieval	✅ Exceeded	Vector retrieval (ChromaDB) rather than the keyword fallback we allowed for
Working LLM reviewer (CWE + remediation)	✅ Met	Two agents, structured JSON
Evidence validation	✅ Met	Two validator bugs found and fixed during evaluation
Connected architecture, stubs documented	✅ Met	§1 above
Benchmark: base repo + ≥4 seeded cases	✅ Exceeded	36 cases (18 pairs), balanced dev/eval split
Initial evaluation round	✅ Exceeded	Five rounds + a final clean run on Shoply (`evaluation-log.md`), plus a held-out real-world run (§3a)
README + setup instructions	✅ Met	`README.md`
Contribution history from all five members	✅ Met	Jia Jun (pipeline, benchmark, evaluation), Kai Zhe (real-world benchmark and evaluation), Jin Rae (GPU model server), Hsu (diff reader hardening), Brian (prompt-injection red-teaming and mitigations)
Progress / scope-change notes	✅ Met	This document
AI-use disclosure log	✅ Met	`ai-use-log.md`, collated by area of work; reproduced in Appendix A

5. Changes to scope, design and technology since the proposal

Change	Why
Benchmark grew from 24 to **36 cases** (added 6 OWASP-Benchmark-derived SQLi pairs, translated from Java to Python/Flask)	Instructor feedback: dataset too small; positives and negatives stay balanced. OWASP Benchmark is Java-only, so we re-implemented its SQLi patterns rather than extracting cases directly

Change	Why
Added a **real-world benchmark**: 12 cases from 6 published CVEs (§3a)	Instructor feedback: test on real projects, not only our own toy app
Held-out set grew from "4 pairs" to a **balanced 9 vulnerable + 9 safe eval split**	Instructor feedback; a fixed dev/eval protocol to stop us tuning on test data
Added a `search_code` tool (semantic RAG over the source)	Instructor feedback: retrieve code, not only docs
ChromaDB chosen over FAISS; embedding model `all-MiniLM-L6-v2`	Persistent local store with metadata (line numbers) for citations
Model is **`qwen2.5-coder:7b`** (proposal: "Qwen via Ollama")	Best local tool-capable model on our hardware; `qwen2.5:7b` was not available
Flask only (proposal: Flask/FastAPI)	One benchmark app was enough; the endpoint extractor still recognises FastAPI routes
LLM-only baseline rebuilt (13 Oct) to match the proposal: no SAST, no RAG, no tools	The first version gave the baseline the same RAG and tools as LGTM, so the comparison could not show what they add. The old setup is kept as the ablation `lgtm-no-sast`
Added "grounded" metrics and an off-category false-alarm metric	Proposal success criterion 3 requires detections *with appropriate evidence*; case-level scoring was hiding authz false alarms on SQLi cases
Added experimental flags `static_facts` and `require_tool_use`	Fixes for failures found in evaluation (see the evaluation log)
Added an optional **GPU-cluster model server** (`model/`, `--ollama-url`)	Laptop runs of the 7B model were slow (about 5 tokens/s in the real-world run), and the cluster lets us try larger models
Hardened the git diff reader	Real repositories contain non-UTF-8 files and paths that git quotes, which crashed reviews
Added **prompt-injection mitigations** behind flags (`--strip-comments`, `--detect-prompt-injection`, both off by default)	Red-teaming showed comments and docstrings in a PR could talk the LGTM agents out of a correct `vulnerable` verdict (§6)

6. Responsible AI: risks and mitigations so far

Risk	Mitigation in LGTM	Residual risk observed
Hallucinated evidence	Evidence validator checks every cited file, line, snippet and policy quote; results are reported with and without ungrounded findings	The model still invents policy file names (e.g. `security_policy.md`) and documentation URLs; the validator cannot check *relevance*, only existence. On real projects, irrelevant but real text (a link sentence, a rule about a different resource) passed validation and was cited as the governing rule
Over-confident verdicts	Four-way verdict scheme including `possible` / `insufficient_evidence`	With context, the model almost never hedges (0 `possible` in LGTM runs)

Risk	Mitigation in LGTM	Residual risk observed
Prompt injection from PR content	PR diff labelled untrusted; tools read base branch only; mitigations strip comments/docstrings before LLM prompt (`strip_comments`) and detect reviewer-directed instructions (`detect_prompt_injection`)	Adversarially tested with 6-case injection suite (see `docs/prompt-injection.md`): unmitigated LGTM was bypassed by comments (downgraded to `possible`) and docstrings (flipped to `safe`); `strip_comments` and injection detector neutralize text-based vectors; adversarial variable renaming remains a residual risk
Privacy / IP	Model runs locally via Ollama, so no code leaves the machine	None for local use
Automation bias	LGTM reports and never auto-fixes; a human decides	Users may still over-trust a confident "safe"
Non-determinism	Temperature 0; identical prompts gave identical verdicts across runs	Verdicts flip when *irrelevant* prompt text changes (a file path in a rule ID flipped one); see the evaluation log

7. Next steps towards the final submission

1. **Prompt-injection residual risks** (Brian): attacks that don't rely on comments, such as identifiers named to look like authorization checks, which stripping and the detector don't stop (`prompt-injection.md` §6).
2. **Field trial on ordinary merged PRs** (Hsu): crash rate, latency and false-alarm rate on real open-source PRs.
3. **Reduce off-category authz false alarms** (open problem in the evaluation log): input scoping or a second-pass verifier, tuned on dev only.
4. **Address the real-world failures** (§3a): run the agent when no endpoint is detected, give it the relevant callees, and test a larger model on the GPU server.
5. **Grow the benchmark**: more OWASP-derived and real-world cases, so one flipped verdict no longer moves F1 by about 0.05.

8. Challenges and what we learned

Each item is a problem we actually hit, what we did about it, and the lesson. Evidence for every one is in the evaluation log, the real-world write-up, the prompt-injection write-up, or the commit history.

1. **Our first headline result was wrong, and the bug was in the evaluator, not the system.** The first full run showed LGTM slightly *worse* than the LLM-only baseline. Two problems caused it: the scorer credited any `vulnerable` finding to the case regardless of category, and the "LLM-only" baseline had quietly been given the same RAG and tools as LGTM, so the comparison could not show what those add. Fixing the evaluation, not the product, flipped the conclusion to F1 0.94 vs 0.75. *Lesson:* audit the metrics code as carefully as the product code, and check every baseline against its written definition line by line.
2. **A silent "wrong version" bug in SAST.** Semgrep was scanning the working tree, not the files at the PR ref, so whichever branch happened to be checked out was what got scanned. Nothing crashed; results were just wrong. We now materialise the PR's files into a temp directory before scanning, with a regression test. *Lesson:* for anything that reads a repository, "which version?" must be explicit.

3. **Aggregate F1 hid our most important failure.** Case-level scoring said everything was fine while the authorization agent was raising ownership alarms on 22 of 24 SQL-injection cases, on endpoints that already had the right check. A developer would experience that as noise on most PRs. Only a per-finding audit exposed it, so we added an off-category alarm metric and a grounded-only F1. *Lesson:* one summary number is not an evaluation; read the individual findings.
4. **Tuning on the test set is easy to do by accident.** Most of the remaining failures were on the eval split, and the natural reflex was to fix the prompt until they passed. We adopted a strict protocol: tune only on dev, fix only genuine bugs motivated by eval cases, and report eval once. Two of three prompt experiments then turned out to be harmful on dev, which we would never have known otherwise. *Lesson:* a held-out split only means something if the process protects it.
5. **A 7B model does not use its tools, and forcing it does not help.** Left alone, `qwen2.5-coder:7b` made zero tool calls across every run, on Shoply and on real projects. When we forced at least one call, it used every tool every time at 6.6× the latency for no accuracy gain. For a small model, pre-retrieving context into the prompt (RAG) works better than letting the model fetch it. *Lesson:* "agentic" is a property of the model as much as the harness; measure tool use, don't assume it.
6. **Determinism is not robustness.** At temperature 0, identical prompts gave identical verdicts, but an absolute file path inside a Semgrep rule ID flipped one verdict and leaked a home directory into the prompt. We made rule IDs path-free and now treat any single-case change as noise. *Lesson:* with 18 eval cases one flip is ±0.05 F1, so report variance and grow the set rather than chase individual cases.
7. **The real-world drop (F1 0.94 → 0.44) was a design limit, not a model limit.** On published CVEs, 7 of 12 LGTM cases ran no agent at all: our pattern-based detectors only fire on route decorators inside the changed hunk, and real fixes put the check far from the decorator or in another file. The bare LLM baseline did better (0.80) partly because the synthetic revert visibly *deletes* the guard. *Lesson:* a benchmark we wrote ourselves encodes our own assumptions (decorator and check in one place); testing on code we did not write is what exposed them.
8. **A benign-looking comment can defeat the reviewer.** Red-teaming showed comments and docstrings in a PR talked the triage agent out of all three SQL-injection detections (100% attack success), while SAST and the tool-free baseline were unaffected. Labelling the diff "untrusted" in the prompt was not enough; stripping comments before the LLM sees the code and flagging reviewer-directed text brought it to 0%. *Lesson:* the more context the model gets, the larger the injection surface, so mitigations must be structural, not just instructions.
9. **Compute was a constant constraint.** The 7B model runs at about 21 s per case on an M-series laptop but about 5 tokens/s on a 4 GB-GPU Windows laptop (68 s per LLM-only case; indexing one large real project took over four minutes). A configured 180 s timeout turned out never to reach the Ollama client, so one runaway generation with no stop token blocked a benchmark run for 76 minutes. We wired the timeout and an output-token cap, built a GPU-cluster model server (`model/`), and always run benchmarks from a frozen copy of the tree so edits made during a multi-hour run cannot contaminate it. *Lesson:* bound every model call, and treat evaluation runs as experiments with provenance.
10. **Real repositories break assumptions a toy app never tests.** Moving off Shoply surfaced non-UTF-8 files and git-quoted paths that crashed the diff reader, a ChromaDB batch limit on large repos, Windows CRLF differences in the case builder, and a Semgrep lookup that failed when the virtualenv was not activated. Each got a regression test. *Lesson:* a field trial on other people's code is the cheapest way to find the engineering bugs an evaluation harness hides.

11. **AI-assisted development is fast, but it needs review discipline.** The AI-generated pipeline worked end to end on day one; the subtle mistakes (items 1 and 2) were in code the unit tests could not catch because the tests encoded the same assumptions. We moved to one PR per piece of work, every number in a report copied from a committed `summary.json`, and a disclosure log by area of work. *Lesson:* use the speed, but keep humans on the claims, the baselines and the metrics.

Appendix A. AI use disclosure log

AI used to build the project, collated by area of work. The LLM that LGTM runs at review time is part of the product and is covered in §6. Tools: **Claude Code** (Anthropic) for most implementation and documentation, **GitHub Copilot** for the GPU model server, and **Google Antigravity** for the prompt-injection red-teaming. All AI-generated code went through a pull request reviewed by a team member before reaching `main`.

#	Dates	Area	Tool	What AI produced	How we verified it
1	10 Oct	Initial implementation	Claude Code	Diff parser, Semgrep runner, RAG index, the two agents, evidence validator, report builder, CLI, the Shoply benchmark app and case generator, test suite	Test suite and full benchmark run; architecture checked against the proposal
2	10 Oct	Proposal feedback	Claude Code	`search_code` tool, dev/eval split, 6 OWASP-derived SQLi pairs	Benchmark re-run; patches regenerated and checked by tests
3	10–13 Oct	Debugging and evaluation fixes	Claude Code	Fixes for SAST scanning the wrong tree, malformed model JSON, the evaluator's category-leak bug, the rebuilt LLM-only baseline; grounded metrics	Regression test per bug; benchmark re-run; report figures copied from `summary.json`, not from the AI
4	13 Oct	GPU model server	GitHub Copilot	Slurm job scripts, deploy scripts, Ollama server helper	Jobs run on the cluster; endpoint tested through the tunnel
5	15 Oct	Real-world CVE benchmark	Claude Code	`cases.json`, `fetch.py`, project policies, `evaluate_real.py`, tests, first draft of the write-up	Advisory ids and SHAs read from the GitHub API, re-verified on every run; all numbers generated by script; each miss read from the case JSON before explaining it
6	17 Oct	Diff reader hardening	Claude Code	UTF-8 decoding with replacement; unquoting of git-quoted paths	New tests in `tests/test_diff.py`; full suite green
7	13–17 Oct	Documentation	Claude Code	Drafts of README, this report, evaluation log, AI use log	Team review; all figures traced to committed run files
8	18 Oct	Prompt-injection red-teaming	Google Antigravity	Threat model, 6-case injection benchmark and generator, evaluation harness, comment stripper and injection detector, Windows Semgrep lookup fix, tests	Full test suite run; Semgrep execution checked on Windows

Decisions made by the team without AI: project scope, the choice of CVEs, the dev/eval protocol, and which results to report. What we learned from working this way is item 11 of §8.